

ILLINOIS TECH | College of Computing

ITMM 485 / 585
Dr. Gurram Gopal

Legal and Ethical Issues in
 Information Technology



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Ethical Concepts and Theories

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**Ch9: Regulating Commerce and
 Speech**
P1: Regulating Cyberspace; DRM



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Announcements

- Internet 02 section- clarification
- Final Exam
- Content on Canvas

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Learning Objectives 1

Upon completion of this lesson the students should be able to:

- Define “regulation by code”
- Describe Digital Rights Management and explain why it is controversial
- Describe spam email and discuss if some forms of spam can be defended
- Explain the difficulties involved in balancing free speech and censorship in online contexts and forums

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Learning Objectives 2

Upon completion of this lesson the students should be able to:

- Describe and discuss issues associated with online distribution of pornography
- Discuss whether online pornography should be censored
- Discuss issues associated what some view as excessive online speech: hate speech as well as speech that can cause physical harm to others
- Define “network neutrality” and explain the implications this principle has for the future of Internet regulation

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Module Objectives

Upon completion of this lesson the students should be able to:

- Define “regulation by code”
- Describe Digital Rights Management and explain why it is controversial

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Regulating Cyberspace

- Should certain forms of speech in cyberspace be regulated?
- If so, which kinds of speech:
 - hate speech?
 - speech that can cause physical harm to others?
 - obscene speech?
- How should those forms of speech be regulated?

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Regulating Cyberspace (Continued)

John Weckert (2007) notes that concerns about cyberspace regulation raise two distinct questions:

1. Should cyberspace be regulated?
2. Can it be regulated?

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Regulating Cyberspace (Continued)

Assuming that cyberspace can be regulated, **who should be responsible** for carrying out the **regulatory functions**:

- the government?
- private organizations?
- Internet users themselves?

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Cyberspace Regulation (Continued)

- 1) What, exactly, do we mean by *cyberspace*?
- 2) What is meant by *regulation* (and, in particular, by "regulation" as it applies to cyberspace)?

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The Ontology of Cyberspace

- Cyberspace is roughly equivalent to the Internet, i.e., the **network of interconnected computers**.
- Is cyberspace a *place*, or is it a *medium* of some sort?
- Is it a virtual space (consisting of all of the data and information that resides in the connected servers and databases on the Internet)?
- Is it a new kind of (broadcast) medium?

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Cyberspace as a Medium

- Godwin (2003) believes that the Internet is a new kind of medium that is significantly different from earlier media, such as the telephone or television.
- The telephone is a "one-to-one medium"
- Television is a "one-to-many medium."
- Goodman suggests that the Internet can be characterized as a "many-to-many medium."



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Cyberspace as a Medium (Continued)

Camp and Chien (2000) note that there are four traditional types of media:

1. *publisher,*
2. *broadcast,*
3. *distributor,*
4. *common carrier.*



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Cyberspace as a Medium (Continued)

- Camp and Chien argue that none of the traditional media models are appropriate for understanding cyberspace.
- An example of a publisher would be a newspaper or a magazine.
- Examples of broadcast media include television and radio.
- Telephone companies and cable companies are examples of common carriers.

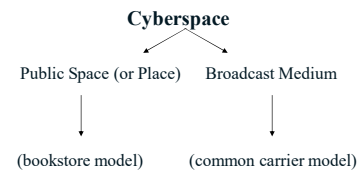
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Cyberspace as a (Public) Place

- Camp and Chien believe that a *spatial model* is a more plausible way to conceive of cyberspace.
- They view cyberspace as a “public space with certain digital characteristics.”
- Note that the model we choose will influence our decisions about how to frame public policies affecting cyberspace regulation.

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Figure 9-1: The Ontology of Cyberspace



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Two Different Senses of "Cyberspace Regulation"

- To "regulate" typically means to **monitor or control a certain product, process, or set of behaviors according to certain requirements, standards, or protocols.**
- Two different senses of "regulation" are used in the debate about whether (and which aspects of) cyberspace should be regulated:
 - 1) regulating **content** (in cyberspace), e.g., online pornography, hate speech, and so forth.
 - 2) regulating **processes** – i.e., rules and policies – for commercial transactions in cyberspace.

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Content vs. Process Based Regulatory Agencies in Physical Space

Content-based :

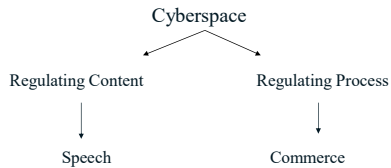
- Food and Drug Administration (FDA);
- Local and State Boards of Health;
- Liquor Control Board.

Process-based :

- Federal Trade Commission (FTC);
- Federal Communications Commission (FCC)
- Security and Exchange Commission (SEC).

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Figure 9-2: Two Modes of Cyberspace Regulation



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Four Modes of Regulating Cyberspace: The Lessig Model

Lawrence Lessig (2000) describes four distinct but interdependent constraints or "modalities," for regulating behavior:

1. *laws*,
2. *social norms*,
3. *market pressures*,
4. *architecture*.



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Analogy: Regulating Smoking Behavior in Physical Space

- Applying the Lessig (fourfold) model to issues affecting smoking, we can use the following schemes:
 - 1) Pass laws against smoking;
 - 2) Apply social pressure (i.e., norms);
 - 3) Apply market pressure (e.g., in pricing practices);
 - 4) Use architecture (e.g., not permitting cigarettes to be sold in vending machines).

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Regulation by Code

- In cyberspace, Lessig notes that *code is law*.
- Software code is analogous to architecture in physical space.
- Lessig notes that many of the earlier computer networks favored *anonymity* (of users).
- He also points out that since the late 1990s, network architectures have favored *control* (i.e., "regulation by code").

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Digital Rights Management (DRM)

- DRM technologies allow content owners to regulate the flow of information in digital media by:
 - blocking access to it via encryption mechanisms, and enabling access to it through the use of passwords.
- The combination of DRM technology and the DMCA (Digital Millennium Copyright Act), described in Chapter 8, make it possible to regulate and enforce policies and laws in cyberspace to a degree that never existed in the physical realm.

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DRM (Continued)

- How is it possible for DRM to enforce such policies?
- Cory Doctorow (2014) points out that there is no copyright law stating that it is illegal to skip through "piracy warnings" when viewing a movie on a DVD player.
- But suppose a viewer wishes to bypass or fast forward through those warnings, as well as through a series of advertisements, displayed prior to the start of the movie.
- Here, the viewer would likely have to break a "digital lock" (on the DVD player) to skip these segments of the DVD.
- But breaking that lock would violate DMCA's anti-circumvention clause (Section 1201 of the DMCA.)

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DRM, Regulation by Code, and Privatization of Information Policy

- Niva Elkin-Koren (2000) worries that because of embedded controls (in code) made possible by DRM, our policies affecting information and digital media are becoming increasingly *privatized*.
- She also notes that this trend toward privatization has enabled software companies to both:
 - a) design code that reflects their own interests and values,
 - b) avoid having to worry about any adverse effects that this can have for the public's interests.

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DRM and Fair-Use Policies

- Pamela Samuelson (2003) notes that DRM technology also allows content owners to exercise far more control over uses of their copyrighted works in digital media than what is provided by conventional copyright law.
- Because of this, she believes that DRM systems may also violate the *fair-use* provision of copyright law.

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DRM and Fair-Use Policies (Cont.)

- Grodzinsky and Bottis (2007) also argue that DRM threatens the fair-use provision of copyright law.
- They believe that because of the way DRM is designed to protect digital content, our conventional understanding of "private use as fair use" has changed.

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Controversial Uses of DRM

- Critics worry about the ways DRM systems can be abused by content owners to control users' computers and to spy on unsuspecting users.
- Some critics also believe that DRM technology has gone too far.
- Review Scenario 9-2 (in the text), describing the "Sony Rootkit controversy" involving a DRM system used by Sony BMG.

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DRM and the Sony Rootkit Controversy

- DRM-related controversies in the Sony BMG case raise the following two questions:
 1. Can users trust content owners, such as Sony BMG, who are easily able to:
 - a) spy on them;
 - b) control aspects of their computers via the use of DRM technology?
 2. Can Sony's practice be justified on the grounds that entertainment companies need DRM systems to protect their intellectual property?

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DRM and the Music Industry

- One controversy involving DRM and the music industry has focused on "interoperability" across the devices on which digital music can be played.
- Interoperability enables users to download and play music on a variety of digital devices.
- But it also challenges the notion that downloadable content can and should be restricted to proprietary devices controlled by the company that owns an "online store," such as iPods in the iTunes store.

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DRM and the Music Industry (Cont.)

- Internationally, there have been some efforts to promote interoperability – e.g., France’s National Assembly passed a law that required distributors of online music in France to remove DRM so that music can be played on any device (Hesseldahl, 2006).
- But many distributors of music content believe that removing DRM to support interoperability could also open the door to file sharing of copyrighted material without compensation for owners.